

5 Cultivation of seaweed for food is well-established in Asia.

- (a) In 2017, the global harvest of seaweed was 32.9 million tonnes. Of this, 96.6 % was cultivated.

Calculate the mass of cultivated seaweed.

Give your answer in standard form.

(1)

Answer..... tonnes

- (b) Seaweeds are photosynthetic organisms.

Photosynthesis can be described as a reaction requiring light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel by combining it with carbon dioxide and releasing oxygen into the atmosphere.

- (i) Which type of bond is broken in a water molecule?

(1)

- ☐ **A** covalent
- ☐ **B** glycosidic
- ☐ **C** hydrogen
- ☐ **D** ionic

- (ii) Which is a fuel that stores hydrogen in plants and seaweeds?

(1)

- ☐ **A** ATP
- ☐ **B** glucose
- ☐ **C** glycogen
- ☐ **D** RUBISCO

(iii) Which row of the table shows the reactions that combine hydrogen with carbon dioxide and release oxygen into the atmosphere?

(1)

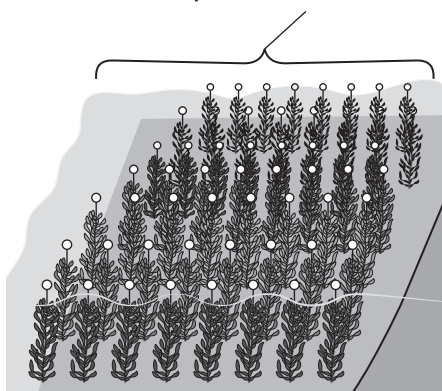
	Hydrogen is combined with carbon dioxide	Oxygen is released into the atmosphere
☒ A	light-dependent reactions	light-dependent reactions
☒ B	light-dependent reactions	light-independent reactions
☒ C	light-independent reactions	light-dependent reactions
☒ D	light-independent reactions	light-independent reactions

*(c) In Europe, cultivation of seaweed takes place on a small scale. Large-scale cultivation of seaweed is now being considered.

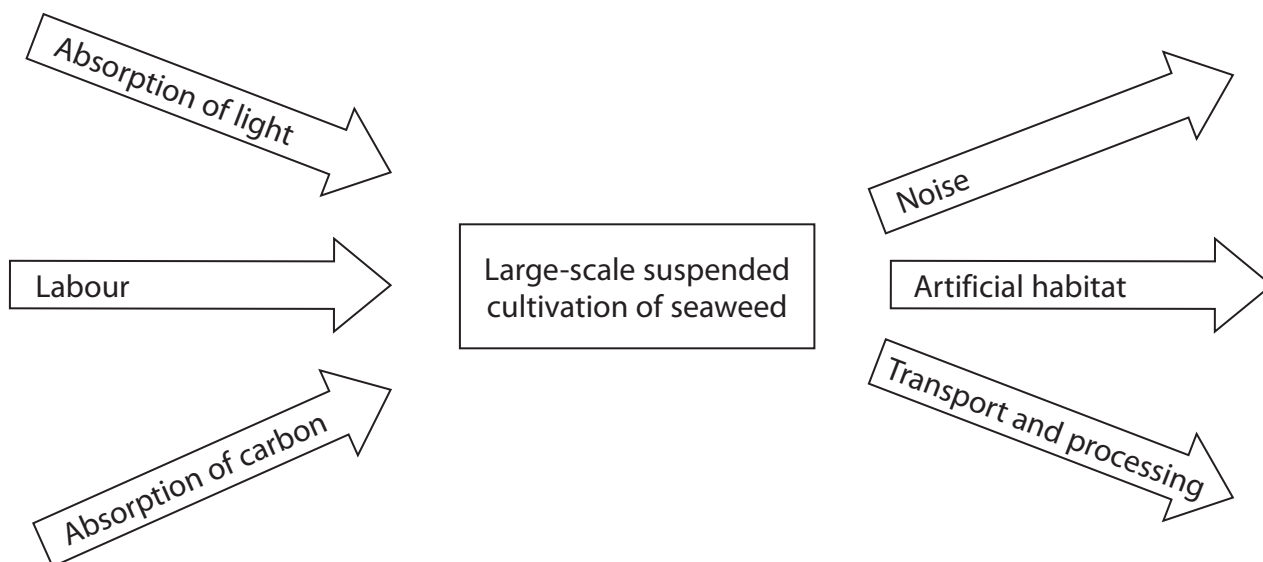
Effective management of the conflict between human needs and conservation is necessary before large-scale cultivation can be used.

The diagram shows seaweed grown by suspended cultivation, a method used in Asia.

Rows of seaweed suspended on wires in the sea



The diagram shows some of the factors that have to be considered when managing the conflict between human needs and conservation.



Use the information in the two diagrams and your own knowledge to support your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(Total for Question 5 = 10 marks)